PIZZA SALES ANALYSIS USING SQL

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Skills: SQL | Data Analysis | Business Insights

Tool Used: MySQL / SQL

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INTRODUCTION

- My name is Geetansh Nangia
- I am currently pursuing Master of Computer Applications (MCA)
- I have a strong interest in Data Analysis and SQL
- I enjoy working with data to extract meaningful insights
- This project reflects my practical skills in analyzing real-world datasets

PROJECT OVERVIEW

- This project is based on analyzing a Pizza Sales Dataset using SQL
- It focuses on extracting useful business insights from raw data
- The project includes basic, intermediate, and advanced SQL queries
- It helps in understanding:
- Sales performance
- Customer preferences
- Revenue trends
- The analysis can support better decision-making in a food business

DATASET OVERVIEW

Pizzas Table

pizza_id	pizza_type_id	size	price
bbq_ckn_s	bbq_ckn	S	12.75
bbq_ckn_m	bbq_ckn	M	16.75
bbq_ckn_l	bbq_ckn	L	20.75
cali_ckn_s	cali_ckn	S	12.75
cali_ckn_m	cali_ckn	M	16.75
cali_ckn_l	cali_ckn	L	20.75
ckn_alfredo_s	ckn_alfredo	S	12.75
ckn_alfredo_m	ckn_alfredo	M	16.75
ckn_alfredo_l	ckn_alfredo	L	20.75
ckn_pesto_s	ckn_pesto	S	12.75
ckn_pesto_m	ckn_pesto	M	16.75
ckn_pesto_l	ckn_pesto	L	20.75
southw_ckn_s	southw_ckn	S	12.75
southw_ckn_m	southw_ckn	M	16.75
southw_ckn_l	southw_ckn	L	20.75
thai_ckn_s	thai_ckn	S	12.75
thai_ckn_m	thai_ckn	M	16.75
thai_ckn_l	thai_ckn	L	20.75
big_meat_s	big_meat	S	12
big_meat_m	big_meat	M	16
big_meat_l	big_meat	L	20.5
classic_dlx_s	classic_dlx	S	12
classic_dlx_m	classic_dlx	M	16
classic_dlx_l	classic_dlx	L	20.5
hawaiian_s	hawaiian	S	10.5
hawaiian_m	hawaiian	M	13.25

Orders Table

er_id	date	time
1	1/1/2015	11:38:36
2	1/1/2015	11:57:40
3	1/1/2015	12:12:28
4	1/1/2015	12:16:31
5	1/1/2015	12:21:30
6	1/1/2015	12:29:36
7	1/1/2015	12:50:37
8	1/1/2015	12:51:37
9	1/1/2015	12:52:01
10	1/1/2015	13:00:15
11	1/1/2015	13:02:59
12	1/1/2015	13:04:41
13	1/1/2015	13:11:55
14	1/1/2015	13:14:19
15	1/1/2015	13:33:00
16	1/1/2015	13:34:07
17	1/1/2015	13:53:00
18	1/1/2015	13:57:08
19	1/1/2015	13:59:09
20	1/1/2015	14:03:08
21	1/1/2015	14:14:29
22	1/1/2015	14:16:26
23	1/1/2015	14:19:03

Order Details Table

r_details_id	order_id	pizza_id	qua
1	1	hawaiian_m	
2	2	classic_dlx_m	
3	2	five_cheese_l	
4	2	ital_supr_l	
5	2	mexicana_m	
6	2	thai_ckn_l	
7	3	ital_supr_m	
8	3	prsc_argla_l	
9	4	ital_supr_m	
10	5	ital_supr_m	
11	6	bbq_ckn_s	
12	6	the_greek_s	
13	7	spinach_supr_s	
14	8	spinach_supr_s	
15	9	classic_dlx_s	
16	9	green_garden_s	
17	9	ital_cpcllo_l	
18	9	ital_supr_l	
19	9	ital_supr_s	
20	9	mexicana_s	
21	9	spicy_ital_l	
22	9	spin_pesto_l	
23	9	veggie_veg_s	

Pizza Types Table

pizza_type_id	name
bbq_ckn	The Barbecue Chi
cali_ckn	The California Chi
ckn_alfredo	The Chicken Alfr
ckn_pesto	The Chicken Pes
southw_ckn	The Southwest Ch
thai_ckn	The Thai Chick
big_meat	The Big Meat
classic_dlx	The Classic Delu
hawaiian	The Hawaiian
ital_cpcllo	The Italian Capoc
napolitana	The Napolitan
pep_msh_pep	The Pepperoni, Mushroom
pepperoni	The Pepperon
the_greek	The Greek P
brie_carre	The Brie Carr
calabrese	The Calabrese
ital_supr	The Italian Supre

SQL ANALYSIS QUERIES OVERVIEW

- Retrieve the total number of orders placed.
- Calculate the total revenue generated from pizza sales.
- Identify the highest-priced pizza.
- Identify the most common pizza size ordered.
- List the top 5 most ordered pizza types along with their quantities.
- Join the necessary tables to find the total quantity of each pizza category ordered.
- Determine the distribution of orders by hour of the day.
- Join relevant tables to find the category-wise distribution of pizzas.
- Group the orders by date and calculate the average number of pizzas ordered per day.
- Determine the top 3 most ordered pizza types based on revenue.
- Calculate the percentage contribution of each pizza type to total revenue.
- Analyze the cumulative revenue generated over time.
- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

1. RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED

```sql
SELECT
    COUNT(order_id) AS total_order_placed
FROM
    orders;
```

OUTPUT

	total_order_placed
▶	21350

2. CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES

```sql
SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales_generate
FROM
    order_details
        JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

OUTPUT

total_sales_generate
817860.05

3. IDENTIFY THE HIGHEST-PRICED PIZZA

```sql
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1
;
```

OUTPUT

	name	price
▶	The Greek Pizza	35.95

4. IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED

```sql
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC
LIMIT 1;
```

OUTPUT

	size	order_count
▶	L	18526

5. LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES

```sql
SELECT
    pizza_types.name AS pizza_name,
    SUM(order_details.quantity) AS quantities
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantities DESC
LIMIT 5;
```

OUTPUT

pizza_name	quantities
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

6. JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED

```sql
SELECT
    pizza_types.category AS category,
    SUM(order_details.quantity) AS total_quantity_ordered
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY category
ORDER BY total_quantity_ordered DESC;
```

OUTPUT

category	total_quantity_ordered
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

7. DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY

```sql
SELECT
    COUNT(orders.order_id) AS order_count,
    HOUR(order_time) AS hours
FROM
    orders
GROUP BY hours;
```

order_count	hours
1231	11
2520	12
2455	13
1472	14
1468	15
1920	16
2336	17
2399	18
2009	19
1642	20
1198	21
663	22
28	23
8	10
1	9

OUTPUT

8. JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS

```sql
SELECT
    COUNT(name), pizza_types.category
FROM
    pizza_types
GROUP BY category;
```

OUTPUT

COUNT(name)	category
6	Chicken
8	Classic
9	Supreme
9	Veggie

9. GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY

```sql
SELECT
    ROUND(AVG(quantity), 0) as avg_pizzas_odered_per_day
FROM
    (SELECT
        orders.order_date AS order_date,
            SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY order_date) AS order_quantity;
```

OUTPUT

avg_pizzas_odered_per_day
138

10. DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE

```sql
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY name
ORDER BY revenue DESC
LIMIT 3;
```

OUTPUT

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

11. CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE

```
SELECT
    pizza_types.category,
    ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
                    ROUND(SUM(order_details.quantity * pizzas.price),
                                2) AS total_sales_generate
                FROM
                    order_details
                        JOIN
                    pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
            2) AS revenue_percentage
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY category
ORDER BY revenue_percentage DESC;
```

OUTPUT

category	revenue_percentage
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

12. ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME

```sql
select order_date , sum(revenue) over (order by order_date)
 as cum_revenue from
(select orders.order_date ,sum(order_details.quantity * pizzas.price)
as revenue
from order_details
join pizzas
on order_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.order_date) as sales ;
```

OUTPUT

order_date	cum_revenue
2015-01-01	2713.850000000
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.350000000
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.300000000
2015-01-14	32358.700000000
2015-01-15	34343.500000000
2015-01-16	36937.650000000
2015-01-17	39001.750000000

13. DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY

```
select name, category,revenue from
(select category, name, revenue, rank()
over(partition by category order by revenue desc) as rn from
(select pizza_types.category , pizza_types.name ,
 sum(order_details.quantity * pizzas.price) as revenue
from pizza_types
join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category , pizza_types.name) as a) as b
where rn<=3;
```

OUTPUT

name	category	revenue
The Thai Chicken Pizza	Chicken	43434.25
The Barbecue Chicken Pizza	Chicken	42768
The California Chicken Pizza	Chicken	41409.5
The Classic Deluxe Pizza	Classic	38180.5
The Hawaiian Pizza	Classic	32273.25
The Pepperoni Pizza	Classic	30161.75
The Spicy Italian Pizza	Supreme	34831.25
The Italian Supreme Pizza	Supreme	33476.75
The Sicilian Pizza	Supreme	30940.5
The Four Cheese Pizza	Veggie	32265.70000000065
The Mexicana Pizza	Veggie	26780.75
The Five Cheese Pizza	Veggie	26066.5

THANK YOU